

Homework — 2.2.2 Computational Methods — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Mark scheme

Part A (14 marks)

1. [1] — Any one: it can be **decomposed** into smaller sub-problems; it has identifiable **inputs, processes and outputs**; it can be solved by a **finite sequence of steps** (an algorithm); it is rule-based/repeatable; the data involved can be **modelled/structured**; useful **abstractions** can be made; it is tractable — solvable in a reasonable time.

2. [2] — 1 mark: **decomposition** breaks a problem into smaller, more manageable sub-problems, which may each be **different** kinds of sub-task. 1 mark: **divide and conquer** repeatedly halves/splits the problem into **smaller instances of the same problem**, solves them, and (where needed) **combines** the results — e.g. binary search, merge sort.

3. - (a) [1] — **Visualisation** (visualisation to solve problems). - (b) [1] — Any one: patterns/trends/outliers are far easier for a human to spot than in raw figures; supports faster decisions; communicates the data more clearly.

4. - (a) [1] — A **heuristic** is a rule of thumb that produces a **good-enough** answer quickly, without guaranteeing the optimal answer. - (b) [2] — 1 mark: an exact method must consider an enormous number of possible route orderings — the number of combinations grows so quickly with the number of stops that the problem becomes **intractable** (cannot be solved exactly in a reasonable time). 1 mark: a heuristic finds a **good-enough** route **quickly**, and a near-optimal route is acceptable in practice.

5. [2] — 1 mark: pipelining splits the process into **stages** that operate **at the same time on different photos** — while photo 1 is being watermarked, photo 2 is being filtered and photo 3 is being resized. 1 mark: because no stage sits idle waiting for a whole photo to finish all three stages, photos are completed at a faster rate — **throughput increases**.

6. - (a) [1] — **Backtracking**. - (b) [2] — 1 mark: the technique builds a solution step by step along one path (a partial solution). 1 mark: when the current path cannot lead to a solution (a dead end), it **undoes the most recent choice(s)** and returns to the last decision point (junction) to try a different option, repeating until the exit is found.

7. [1] — **Data mining**.

Part B (6 marks)

8. [1] — A **function returns a value** (so a call can be used in an expression); a procedure does not return a value.

9. [2] — 1 mark: a **stack overflow**. 1 mark: with no reachable base case the subroutine keeps calling itself; each call pushes a new frame (return address/parameters/locals) onto the **call stack** until the stack's memory is exhausted.

10. [3] — 1 mark: **thinking ahead** is identifying in advance what a solution will need — its inputs and outputs, preconditions, and reusable components — before/while designing it. 1 mark per distinct example, max 2: deciding what data a form must capture before building it; **caching** data/resources likely to be needed again; specifying **preconditions** for a subroutine; planning reusable components/subroutines.

Total: 14 + 6 = 20 marks